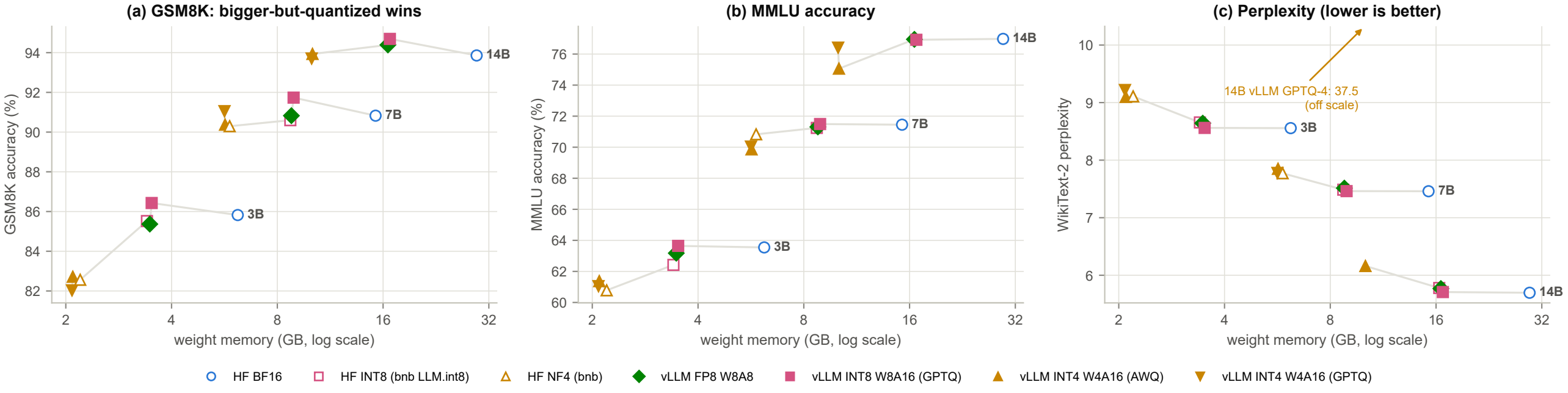

(a) GSM8K: bigger-but-quantized wins
GSM8K accuracy (%)
94
92
90
88
86
84
82
14B
7B
3B
2
4
8
16
32
weight memory (GB, log scale)
(b) MMLU accuracy
MMLU accuracy (%)
76
74
72
70
68
66
64
62
60
14B
7B
3B
2
4
8
16
32
weight memory (GB, log scale)
(c) Perplexity (lower is better)
WikiText-2 perplexity
10
9
8
7
6
14B vLLM GPTQ-4: 37.5
(off scale)
3B
7B
14B
2
4
8
16
32
weight memory (GB, log scale)
HF BF16
HF INT8 (bnb LLM.int8)
HF NF4 (bnb)
vLLM FP8 W8A8
vLLM INT8 W8A16 (GPTQ)
vLLM INT4 W4A16 (AWQ)
vLLM INT4 W4A16 (GPTQ)